Pandas Cheat Sheet

The essential commands for every data science project

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01 LOADING & EXPLORING DATA

IMPORT & LOAD

```
import pandas as pd
df = pd.read_csv("file.csv")                          # load CSV
df = pd.read_excel("file.xlsx")                       # load Excel
df = pd.read_json("file.json")                        # load JSON
df = pd.read_csv("file.csv", usecols=["col1","col2"])
```

FIRST LOOK — RUN THESE BEFORE ANYTHING ELSE

```
df.head()                                             # first 5 rows
df.tail()                                             # last 5 rows
df.shape                                              # (rows, cols)
df.columns                                            # column names
len(df)                                               # row count
df.info()                                             # types + nulls
df.describe()                                         # stats summary
df.dtypes                                             # column types
df.nunique()                                          # unique per col
df.sample(5)                                          # random 5 rows
```

02 CLEANING — NULLS, DUPLICATES & TYPES

NULL VALUES

```
df.isnull().sum()                                     # count
df.dropna()                                           # remove
df.dropna(subset=["col"])                             # one col
df.fillna(0)                                          # fill 0
df["col"].fillna(df["col"].mean())
```

TYPES & DUPLICATES

```
df.duplicated().sum()                                 # count
df.drop_duplicates()                                  # remove
df["col"].astype(int)                                 # to int
df["col"].astype(str)                                 # to str
pd.to_datetime(df["date"])                            # to date
```

03 FILTERING & SELECTING

SELECT COLUMNS

```
df["col"]                                             # 1 col
df[["col1","col2"]]                                   # multi
df.iloc[0:5]                                          # by index
df.loc[0, "col"]                                      # row+col
```

FILTER ROWS

```
df[df["age"] > 30]
df[df["city"]=="Paris"]
df[df["col"].isin(list)]
df.query("age > 30")
```

MULTIPLE CONDITIONS

```
df[(df["age"] > 30) & (df["city"] == "Paris")]        # AND
df[(df["age"] < 20) | (df["age"] > 60)]               # OR
df[~df["col"].isnull()]                               # NOT null
```

04 GROUPING & AGGREGATING

BASIC AGGREGATIONS

```
df.groupby("category")["sales"].mean()                # mean per group
df.groupby("category")["sales"].sum()                 # sum per group
df.groupby("category")["sales"].count()               # count per group
df.groupby("category")["sales"].agg(["mean","sum","count"])
```

MULTIPLE AGGREGATIONS

```
df.groupby("cat").agg(
    mean_s=("sales","mean"),
    total=("sales","sum"),
    n=("sales","count")
)
```

GROUP BY MULTIPLE COLS

```
df.groupby(
    ["region","category"]
)["sales"].sum()
# Reset index after groupby
```

05 SORTING & RANKING

SORT

```
df.sort_values("col")                                 # asc
df.sort_values("col", ascending=False)                # desc
df.sort_values(["c1","c2"])                           # multi
df.sort_index()                                       # by idx
```

TOP / BOTTOM

```
df.nlargest(5,"sales")                                # top 5
df.nsmallest(5,"sales")                               # bottom 5
df["col"].rank()                                      # rank col
df["col"].value_counts()                              # freq tbl
df["col"].unique()                                    # unique vals
```

06 MERGING & JOINING

```
pd.merge(df1, df2, on="id")                           # inner join (default)
pd.merge(df1, df2, on="id", how="left")               # left join
pd.merge(df1, df2, on="id", how="right")              # right join
pd.merge(df1, df2, on="id", how="outer")              # outer join
pd.merge(df1, df2, left_on="id", right_on="uid")      # diff col names
pd.concat([df1, df2], ignore_index=True)              # stack rows
pd.concat([df1, df2], axis=1)                         # stack columns
```

QUICK TIPS

`df.info()` Run df.info() first — always catch type and null issues before analysis	`df.copy()` Use df.copy() before cleaning — never overwrite raw data	`df.reset_index(drop=True)` After filtering to fix the index
`df.rename(columns={"old":"new"})` To rename columns cleanly	`df["col"].str.lower().str.strip()` To clean messy text columns	`df.memory_usage(deep=True)` To check dataframe size on large datasets

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